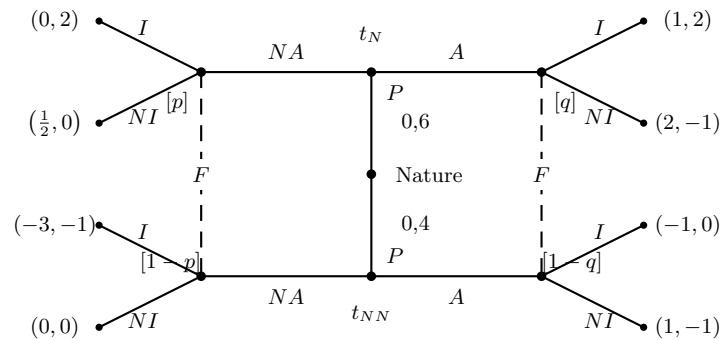


Signalling game

Consider the following game between Phoenicia F and Persia P . Nature first chooses Persia's type. With probability $0,6 = \frac{3}{5}$, Persia has nuclear weapons, t_N . With probability $0,4 = \frac{2}{5}$, Persia does not have nuclear weapons, t_{NN} . Persia observes its own type and chooses whether to attack, A , or not attack, NA . Phoenicia observes Persia's action, but not Persia's type, and chooses whether to invade, I , or not invade, NI . Payoffs are ordered as (P, F) .



1. Explain what p and q represent in this context
2. Can there be a Perfect Bayesian Nash equilibrium in which both types of P do not attack? If so, write F 's response in each case and the beliefs that sustain it. If not, explain why there are incentives to deviate
3. Can there be a Perfect Bayesian Nash equilibrium in which each type of P reveals itself, that is, t_N attacks and t_{NN} does not attack? If so, write F 's response and the beliefs that sustain it. If not, explain why there are incentives to deviate

Solution

1. The beliefs p and q are Phoenicia's posterior probabilities that Persia has nuclear weapons after observing Persia's action:

$$p = \Pr(t_N \mid NA) \quad q = \Pr(t_N \mid A)$$

Thus, p is the probability that Persia is nuclear after Phoenicia observes NA , and q is the probability that Persia is nuclear after Phoenicia observes A .

Conclusion: p and q are Phoenicia's posterior beliefs about Persia's type after observing NA and A , respectively

2. First, consider Phoenicia's optimal response after observing NA . If Phoenicia believes that Persia is nuclear with probability p , then

$$U_F(I \mid NA) = 2p + (-1)(1 - p) = 3p - 1 \quad U_F(NI \mid NA) = 0$$

Therefore,

$$I \text{ is optimal after } NA \iff 3p - 1 \geq 0 \iff p \geq \frac{1}{3} \approx 0,333$$

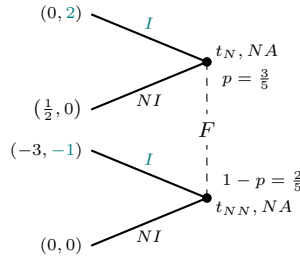
Now suppose both types of Persia choose NA . Then, by Bayes' rule,

$$p = \Pr(t_N \mid NA) = \frac{3}{5} = 0,6$$

Since

$$\frac{3}{5} = 0,6 > \frac{1}{3} \approx 0,333$$

Phoenicia chooses I after observing NA .



After observing A , if Phoenicia believes that Persia is nuclear with probability q , then

$$U_F(I \mid A) = 2q + 0(1 - q) = 2q \quad U_F(NI \mid A) = (-1)q + (-1)(1 - q) = -1$$

Since $2q \geq -1$ for every $q \in [0, 1]$, Phoenicia chooses I after observing A for any off-path belief q .

Given these responses, the nuclear type obtains

$$u_P(NA, I \mid t_N) = 0 \quad u_P(A, I \mid t_N) = 1$$

Thus, the nuclear type wants to deviate to A . The non-nuclear type obtains

$$u_P(NA, I \mid t_{NN}) = -3 \quad u_P(A, I \mid t_{NN}) = -1$$

Thus, the non-nuclear type also wants to deviate to A .

Conclusion: there is no Perfect Bayesian Nash equilibrium in which both types of Persia choose NA . If both types choose NA , then $p = \frac{3}{5} = 0,6$, so Phoenicia chooses I after NA , and both types of Persia have an incentive to deviate to A

3. Now consider the separating strategy profile in which each type reveals itself:

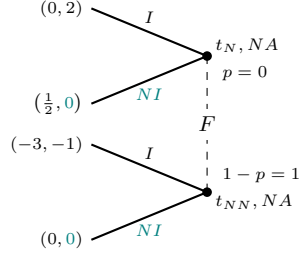
$$t_N \rightarrow A \quad t_{NN} \rightarrow NA$$

By Bayes' rule,

$$q = \Pr(t_N | A) = 1 \quad p = \Pr(t_N | NA) = 0$$

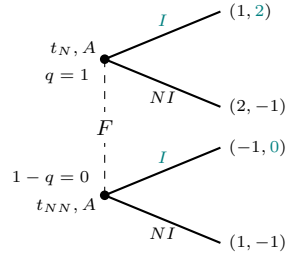
Given $p = 0$, after observing NA , Phoenicia chooses NI , because

$$U_F(I | NA) = -1 \quad U_F(NI | NA) = 0$$



Given $q = 1$, after observing A , Phoenicia chooses I , because

$$U_F(I | A) = 2 \quad U_F(NI | A) = -1$$



Now check Persia's incentives. For the nuclear type,

$$u_P(A, I | t_N) = 1 \quad u_P(NA, NI | t_N) = \frac{1}{2} = 0,5$$

Thus, the nuclear type does not want to deviate. For the non-nuclear type,

$$u_P(NA, NI | t_{NN}) = 0 \quad u_P(A, I | t_{NN}) = -1$$

Thus, the non-nuclear type does not want to deviate.

Conclusion: there is a separating Perfect Bayesian Nash equilibrium in which

$$t_N \rightarrow A \quad t_{NN} \rightarrow NA \quad F(A) = I \quad F(NA) = NI$$

with beliefs

$$q = 1 \quad p = 0$$

In this equilibrium, Persia's action fully reveals its type